

MA 214 Week 1: Lab 1

Working with R and Data

1. **Your name:** _____

BU ID: _____

2. **Partner's name:** _____

BU ID: _____

Big idea: Data analysis starts with understanding what each row and column represents before calculating summaries or making plots. **Do not use GenAI tools for this lab.** The point is to practice your own analysis work, coding skills, and statistical reasoning.

Files used: `flights.csv` and `lab-starter.R`

Skills practiced: `read.csv()`, `glimpse()`, `count()`, `summarize()`, and basic `ggplot()` graphics.

Your mission: Use the `flights` data to practice importing, inspecting, summarizing, and visualizing data in a reproducible R script.

Icebreaker

Before opening R: introduce yourself to your partner and answer the warm-up question below.

Warm-up response: What is one everyday data set you have seen recently, and what would one row represent in that data set?

Getting Started

Open `lab-starter.R`. Make sure `flights.csv` is in the same folder as your script. Do not use `setwd()` in the file you submit.

```
library(dplyr)
library(ggplot2)
library(readr)

flights <- read.csv("flights.csv")

head(flights)
str(flights)
```

HUDDLE UP

Before calculating anything, what do you think one row represents? Which variables seem useful for describing where flights went and how long they took?

Question 1. Explore the Flights Data

Start by getting oriented. Record what you learn about the size, columns, and variable types in the data.

```
dim(flights)
names(flights)
str(flights)
```

Variable or data set	Type or role	What it means
origin		
dest		
air_time		

Write 2-3 sentences describing what you learned from the first look at the data.

Question 2. Make a Prediction

Before summarizing, predict which NYC airport had the most flights in this data set: JFK, LGA, or EWR.

PREDICTION TIME

Before running the code, predict the busiest origin airport. Be specific enough that the data can support or challenge your answer.

```
airport_counts <- flights %>%
  count(origin)

airport_counts
```

What did your team predict? What evidence did you check first?

Question 3. Compute Summaries

Use counts and grouped summaries to answer concrete questions about destinations and air time.

```
state_counts <- flights %>%  
  count(dest)  
  
air_time_by_month <- flights %>%  
  group_by(month) %>%  
  summarize(  
    mean_air_time = mean(air_time, na.rm = TRUE),  
    n_flights = n()  
  )
```

HUDDLE UP

Does each summary answer the question directly, or do you need another comparison, plot, or table?

CLAIM, EVIDENCE, CONTEXT

The main pattern is _____.

My evidence is _____.

This matters because _____.

Question 4. Interpret in Context

Connect your R output to a plain-language statement about the flights data.

Question	Your answer
What comparison did you make?	
What result is strongest?	
What limitation or caution should readers know?	

DEBATE CORNER

If you and your partner disagree about the interpretation, write both arguments first. Then decide what claim the evidence actually supports.

Question 5. Close the Lab

Submit your completed `lab-starter.R`. Your script should run from top to bottom and include the requested objects and plots.

Final response: state one claim about the flights data, cite one piece of evidence, and explain the context.

Optional Challenge

Make one plot that compares `air_time` across months or origin airports. Save it using `ggsave()`.

```
# Optional challenge starter code.  
ggplot(flights, aes(x = factor(month), y = air_time)) +  
  geom_boxplot()
```